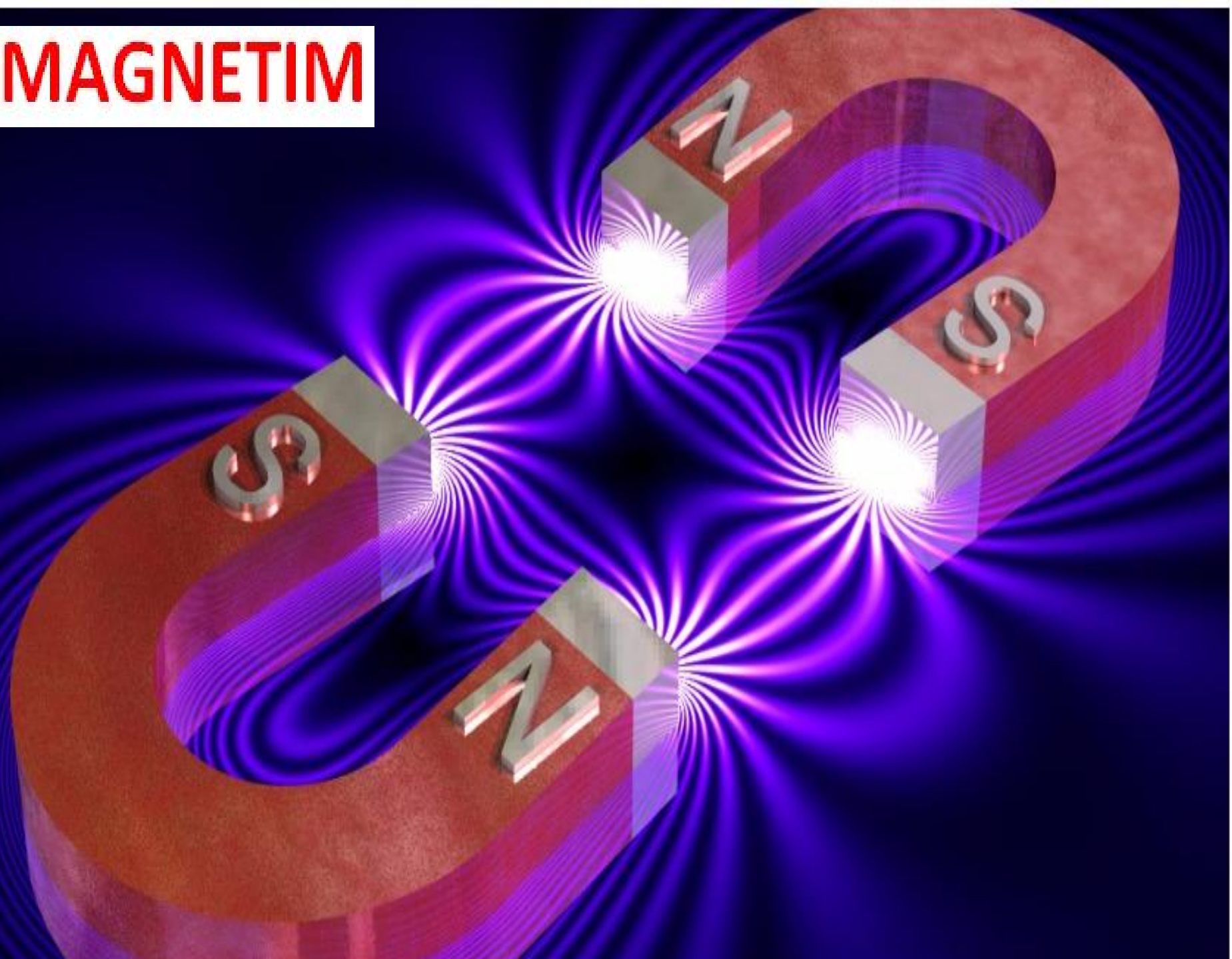


# MAGNETIM



## 2.1. MAGNETIC FORCE AND MAGNETIC FIELD

### 2.1.1. Magnetic field lines

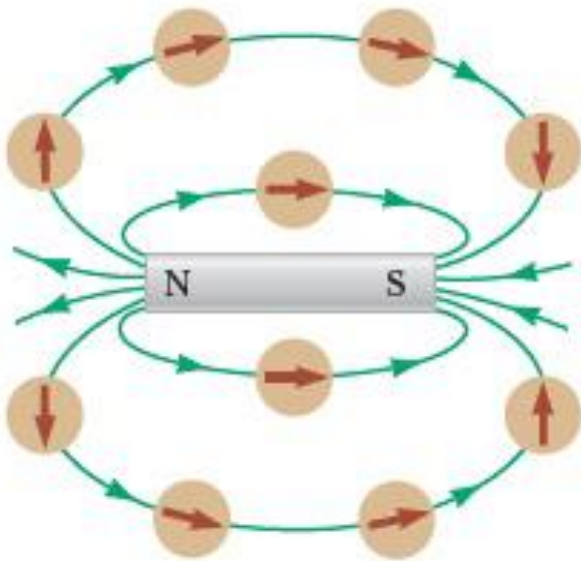
*Magnetism* is a class of physical phenomena that are mediated by magnetic fields. Electric currents and the magnetic moments of elementary particles give rise to a magnetic field, which acts on other currents and magnetic moments.

In our study of electricity, we described the interactions between charged objects in terms of electric fields. Recall that an electric field surrounds any electric charge. In addition to containing an electric field, the region of space surrounding any *moving* electric charge also contains a **magnetic field**. A magnetic field also surrounds a magnetic substance making up a permanent magnet.

# Magnetic field lines

Historically, the symbol  $\mathbf{B}$  has been used to represent a magnetic field, and we use this notation in this chapter. The direction of the magnetic field  $\vec{\mathbf{B}}$  at any location is the direction in which a compass needle points at that location. As with the electric field, we can represent the magnetic field by means of drawings with *magnetic field lines*.

Compass needles can be used to trace the magnetic field lines in the region outside a bar magnet.



- The figure shows how the magnetic field lines of a bar magnet can be traced with the aid of a compass
- Notice that the magnetic field lines outside the magnet point away from the north pole and toward the south pole

# Magnetic field at a point in space

We can quantify the magnetic field  $\vec{B}$  by using our model of a particle in a field, like the model discussed for gravity and for electricity. The existence of a magnetic field at some point in space can be determined by measuring the **magnetic force**  $\vec{F}_B$  exerted on an appropriate test particle placed at that point. This process is the same one we followed in defining the electric field in previous chapter. If we perform such an experiment by placing a particle with charge  $q$  in the magnetic field, we find the following results that are similar to those for experiments on electric forces:

# Magnetic field at a point in space

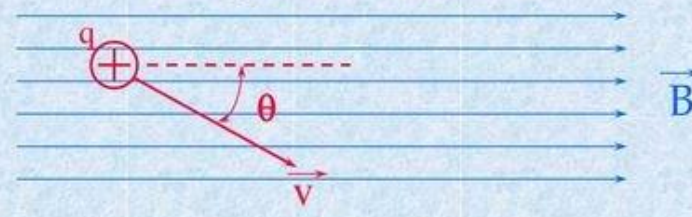
- The magnetic force is proportional to the charge  $q$  of the particle.
- The magnetic force on a negative charge is directed opposite to the force on a positive charge moving in the same direction.
- The magnetic force is proportional to the magnitude of the magnetic field vector  $\vec{B}$

## Magnetic field at a point in space

We also find the following results, which are *totally different* from those for experiments on electric forces:

- The magnetic force is proportional to the speed  $v$  of the particle.
- If the velocity vector makes an angle with the magnetic field, the magnitude of the magnetic force is proportional to sine of that angle
- When a charged particle moves *parallel* to the magnetic field vector, the magnetic force on the charge is zero.

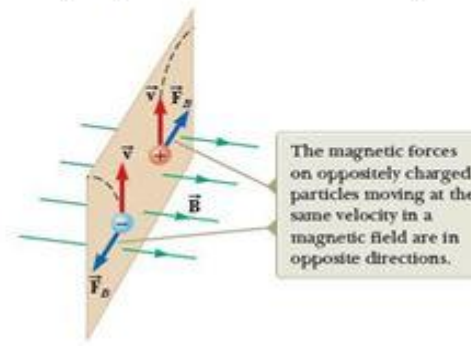
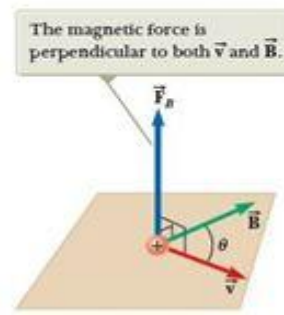
**Magnetic Fields**



- ❖ The magnitude of the magnetic force,  $\vec{F}_B$ , is proportional to the charge  $q$ , the speed  $v = |\vec{v}|$  of the particle, and the magnetic field  $\vec{B}$ .
- ❖  $|\vec{F}_B|$  also depends on  $\theta$ .

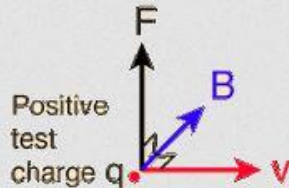
$$F_B = |\vec{F}_B| = |q|vB \sin \theta$$

- When a charged particle moves in a direction **not parallel** to the magnetic field vector, the magnetic force acts in a direction perpendicular to both  $\vec{v}$  and  $\vec{B}$  that is, the magnetic force is perpendicular to the plane formed by  $\vec{v}$  and  $\vec{B}$



# Magnetic field at a point in space

- ✓ Despite this complicated behavior, these observations can be summarized in a compact way by writing the magnetic force in the form

$$\vec{F} = q\vec{v} \times \vec{B}$$


Positive test charge  $q$ .

- ✓ By using the properties of cross product, the magnitude of magnetic force on a charged particle is

$$F_{\text{mag}} = |\mathbf{F}_{\text{mag}}| = q |\mathbf{v}| |\mathbf{B}| \sin \theta = qvB \sin \theta$$

where  $\theta$  is the angle between the velocity vector  $\mathbf{v}$  and the magnetic field vector  $\mathbf{B}$  at this point in space

- ✓ From Equation to the right side, we see that the SI unit of magnetic field is the Newton per coulomb-meter per second, which is called the tesla (T):

$$1 \text{ T} = \frac{\text{N}}{\text{C} \cdot \text{m/s}}$$

Because a coulomb per second is defined to be an ampere, we see that:

$$1 \text{ T} = 1 \frac{\text{N}}{\text{A} \cdot \text{m}}$$

## Let's compare the important differences between the electric and magnetic versions of the particle in a field model:

- The electric force vector is along the direction of the electric field, whereas the magnetic force vector is perpendicular to the magnetic field.
- The electric force acts on a charged particle regardless of whether the particle is moving, whereas the magnetic force acts on a charged particle only when the particle is in motion.
- The electric force does work in displacing a charged particle, whereas the magnetic force associated with a steady magnetic field does no work when a particle is displaced because the force is perpendicular to the displacement of its point of application.

*From the last statement and on the basis of the work–kinetic energy theorem, we conclude that the kinetic energy of a charged particle moving through a magnetic field cannot be altered by the magnetic field alone. The field can alter the direction of the velocity vector, but it cannot change the speed or kinetic energy of the particle.*

## 2.1.2. Motion of a Charged Particle in a Uniform Magnetic Field

We found that the magnetic force acting on a charged particle moving in a magnetic field is perpendicular to the velocity of the particle and that consequently the work done on the particle by the magnetic force is zero. Let us now consider the special case of a positively charged particle moving in a uniform magnetic field with the initial velocity vector of the particle perpendicular to the field. Let us assume that the direction of the magnetic field is into the page. Figure 2.5 shows that the particle moves in a circle in a plane perpendicular to the magnetic field. The particle moves in this way because the magnetic force  $\vec{F}_B$  is at right angles to  $\vec{v}$  and  $\vec{B}$  and has a constant magnitude  $qvB$ . As the force deflects the particle, the directions of  $\vec{v}$  and  $\vec{F}_B$  change continuously, as Figure 2.5 shows. Because  $\vec{F}_B$  always points toward the center of the circle, it changes only the direction of  $\vec{v}$  and not its magnitude.

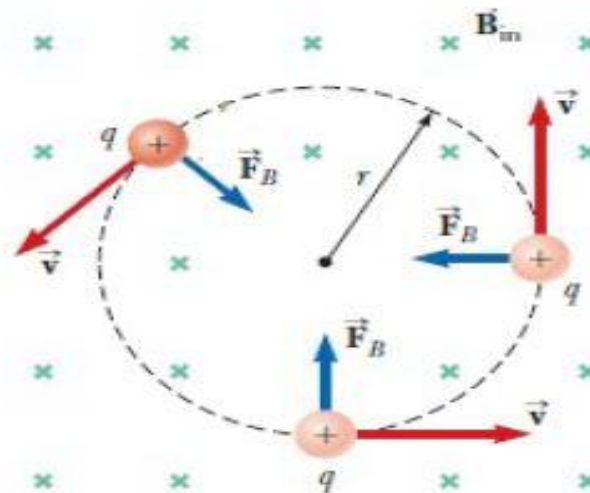


Figure 2.5: *Circulation of a charged particle in a uniform magnetic field*

# Motion of a Charged Particle in a Uniform Magnetic Field

As Figure 2.5 illustrates, the rotation is counter-clockwise for a positive charge. If  $q$  were negative, the rotation would be clockwise. We can use the second Newton's law to equate this magnetic force to the radial force required to keep the charge moving in a circle:

$$\sum F = ma$$

Because the particle moves in a circle, we also model it as a particle in uniform circular motion and we replace the acceleration with centripetal acceleration:

$$F_B = qvB = m\frac{v^2}{r} \quad (2.1.4)$$

This expression leads to the following equation for the radius of the circular path:

$$r = \frac{mv}{|q|B} \quad (2.1.5)$$

# Motion of a Charged Particle in a Uniform Magnetic Field

That is, the radius of the path is proportional to the linear momentum  $mv$  of the particle and inversely proportional to the magnitude of the charge on the particle and to the magnitude of the magnetic field. The angular speed of the particle is

$$\omega = \frac{v}{r} = \frac{|q|B}{m} \quad (2.1.6)$$

The period of the motion (the time interval the particle requires to complete one revolution) is equal to the circumference of the circle divided by the speed of the particle:

$$T = \frac{2\pi r}{v} = \frac{2\pi}{\omega} = \frac{2\pi m}{|q|B} \quad (2.1.7)$$

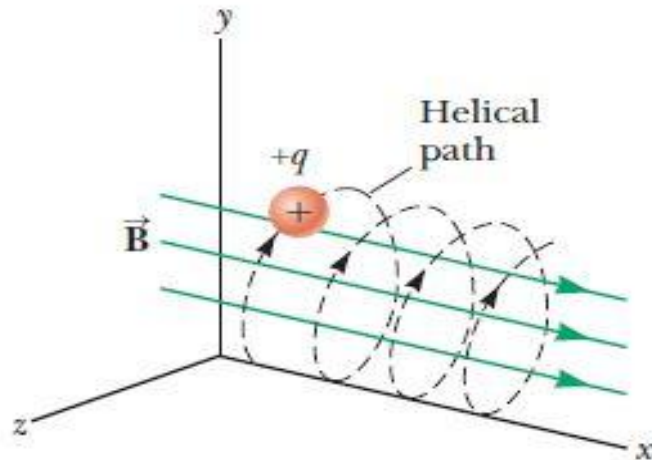
These results shows that the angular speed of the particle and the period of the circular motion do not depend on the linear speed of the particle or on the radius of the orbit. The angular speed  $\omega$  is often referred to as cyclotron frequency because charged particles circulate at this angular speed in the type of accelerator called a cyclotron.

If a charged particle moves in a uniform magnetic field with its velocity at some arbitrary angle with respect to  $\vec{B}$ , its path is a helix.

# Motion of a Charged Particle in a Uniform Magnetic Field

If a charged particle moves in a uniform magnetic field with its velocity at some arbitrary angle with respect to  $\vec{\mathbf{B}}$ , its path is a helix. For example, if the field is directed in the  $x$  direction as shown in Figure 29.9, there is no component of force in the  $x$  direction. As a result,  $a_x = 0$ , and the  $x$  component of velocity remains constant. The charged particle is in equilibrium in this direction. The magnetic force  $q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$  causes the components  $v_y$  and  $v_z$  to change in time, however, and the resulting motion is a helix whose axis is parallel to the magnetic field. The projection of the path onto the  $yz$  plane (viewed along the  $x$  axis) is a circle. (The projections of the path onto the  $xy$  and  $xz$  planes are sinusoids!) Equations 2.1.5 to 2.1.7 still apply provided  $v$  is replaced by  $v_{\perp} = \sqrt{v_y^2 + v_z^2}$ .

**Figure 29.9** A charged particle having a velocity vector that has a component parallel to a uniform magnetic field moves in a helical path.



# Lorentz Force Law

We can define the magnetic field **B** (a vector quantity) at a point by the vector force **F<sub>mag</sub>** at that point experienced by a particle with charge **q** and velocity **v**:

$$\mathbf{F}_{\text{mag}} = q \mathbf{v} \times \mathbf{B}$$

If there is also an electric field **E** at this point, then in addition to the above magnetic force, there will be an electric force **F<sub>elec</sub> = qE** and the total force **F<sub>tot</sub>** on the charge will be

$$\begin{aligned} \mathbf{F}_{\text{tot}} &= \mathbf{F}_{\text{elec}} + \mathbf{F}_{\text{mag}} = q \mathbf{E} + q \mathbf{v} \times \mathbf{B} \\ &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \end{aligned}$$

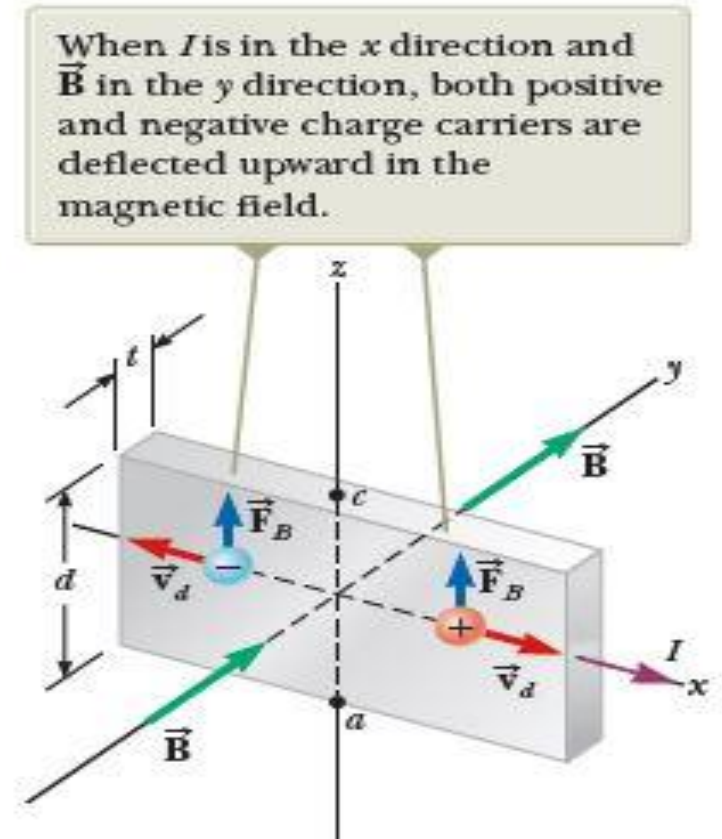
## 2.1.3. The Hall effect

When a current-carrying conductor is placed in a magnetic field, a potential difference is generated in a direction perpendicular to both the current and the magnetic field. This phenomenon, first observed by Edwin Hall in 1879, is known as the *Hall Effect*.

The arrangement for observing the Hall effect consists of a flat conductor carrying a current  $I$  in the  $x$  direction as shown in Figure 29.26. A uniform magnetic field  $\vec{B}$  is applied in the  $y$  **direction**. If the charge carriers are electrons moving in the **negative  $x$  direction** with a drift velocity  $\vec{v}_d$  they experience an upward magnetic force  $\vec{F}_B = q\vec{v}_d \times \vec{B}$ , are deflected upward, and accumulate at the upper edge of the flat conductor, leaving an excess of positive charge at the lower edge. This accumulation of charge at the edges establishes an electric field in the conductor and increases until the electric force on carriers remaining in the bulk of the conductor balances the magnetic force acting on the carriers.

# The Hall effect

The electrons can now be described by the particle in equilibrium model, and they are no longer deflected upward. A sensitive voltmeter connected across the sample can measure the potential difference, known as the **Hall voltage**, generated across the conductor  $\Delta V_H$ :



**Figure 29.26** To observe the Hall effect, a magnetic field is applied to a current-carrying conductor. The Hall voltage is measured between points  $a$  and  $c$ .

# The Hall effect

In deriving an expression for the Hall voltage, first note that the magnetic force exerted on the carriers has magnitude  $qv_dB$ . In equilibrium, this force is balanced by the electric force  $qE_H$ , where  $E_H$  is the magnitude of the electric field due to the charge separation (sometimes referred to as the *Hall field*). Therefore,

$$qv_dB = qE_H$$

$$E_H = v_dB$$

If  $d$  is the width of the conductor, the Hall voltage is

$$\Delta V_H = E_H d = v_d B d \quad (29.19)$$

## 2.1.4 Magnetic Force acting on a Current-carrying conductor

Let us quantify this discussion by considering a straight segment of wire of length  $L$  and cross-sectional area  $A$ , carrying a current  $I$  in a uniform magnetic field  $\vec{B}$ , as shown in figure 2.9. The magnetic force exerted on a charge  $q$  moving with a drift velocity  $\vec{v}_d$  is  $q\vec{v} \times \vec{B}$ . To find the total force acting on the wire, we multiply the force exerted on one charge by the number of charges in the segment. Because the volume of the segment is  $AL$ , the number of charges in the segment is  $nAL$ , where  $n$  is the number of charges per unit volume.

Hence, the total magnetic force on the wire of length  $L$  is

$$\vec{F} = q(\vec{v}_d \times \vec{B})nAL \quad (2.1.10)$$

We can write this expression in a more convenient form by noting that, from the fact that the current in the wire is  $I = nqv_dA$ . Therefore,

$$\vec{F}_B = I \vec{L} \times \vec{B} \quad (2.1.11)$$

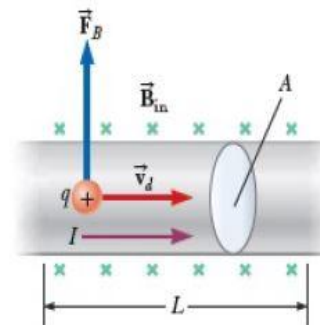


Figure 2.9: A segment of a current-carrying wire located in a magnetic field  $\vec{B}$

where  $\vec{L}$  is a vector that points in the direction of the current  $I$  and has a magnitude equal to the length  $L$  of the segment. This expression applies only to a straight segment of wire in a uniform magnetic field.

Now consider an arbitrarily shaped wire segment of uniform cross section in a magnetic field as shown in Figure 2.10. It follows from Equation 2.1.11 that the magnetic force exerted on a small segment of vector length  $d\vec{s}$  in the presence of a field  $\vec{B}$  is

$$d\vec{F}_B = I d\vec{s} \times \vec{B} \quad (2.1.12)$$

where  $d\vec{F}_B$  is directed out of the page for the directions of  $\vec{B}$  and  $d\vec{s}$  in Figure 2.10. Equation 2.1.12 can be considered as an alternative definition of  $\vec{B}$ . That is, we can define the magnetic field  $\vec{B}$  in terms of a measurable force exerted on a current element, where the force is a maximum when  $\vec{B}$  is perpendicular to the element and zero when  $\vec{B}$  is parallel to the element.

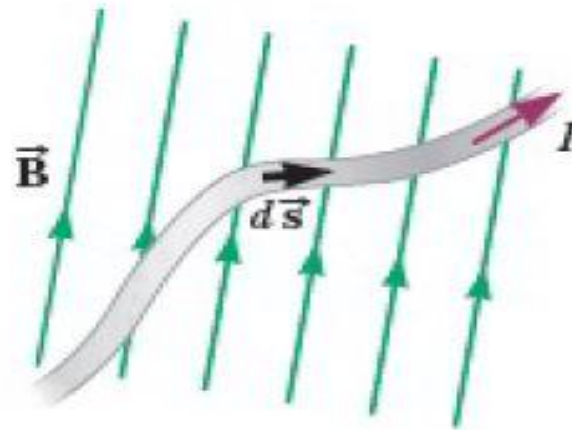


Figure 2.10: A wire segment of arbitrary shape carrying a current  $I$  in a magnetic field  $\vec{B}$  experiences a magnetic force.

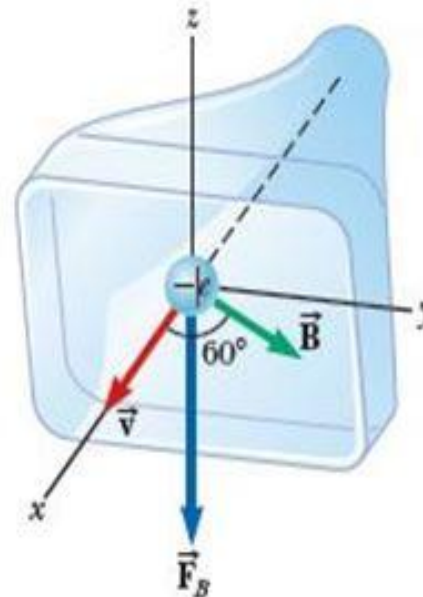
To calculate the total force  $\vec{F}_B$  acting on the wire shown in Figure 2.10, we integrate Equation 2.1.12 over the length of the wire:

$$\vec{F}_B = I \int_a^b d\vec{s} \times \vec{B} \quad (2.1.13)$$

where  $a$  and  $b$  represent the end points of the wire. When this integration is carried out, the magnitude of the magnetic field and the direction the field makes with the vector  $d\vec{s}$  may differ at different points.

# Example 1

An electron in an old-style television picture tube moves toward the front of the tube with a speed of  $8.0 \times 10^6 \text{ m/s}$  along the  $x$  axis (Figure below). Surrounding the neck of the tube are coils of wire that create a magnetic field of magnitude  $0.025 \text{ T}$ , directed at an angle of  $60^\circ$  to the  $x$  axis and lying in the  $xy$  plane.



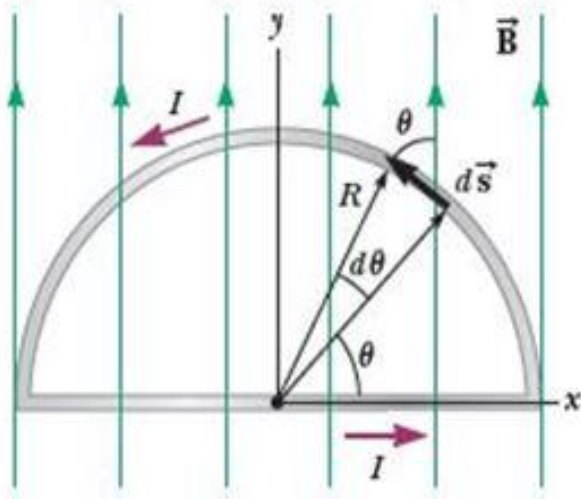
Calculate the magnetic force on the electron.

## Example 2

A proton is moving in a circular orbit of radius 14 cm in a uniform 0.35-T magnetic field perpendicular to the velocity of the proton. Find the speed of the proton.

## Example 3

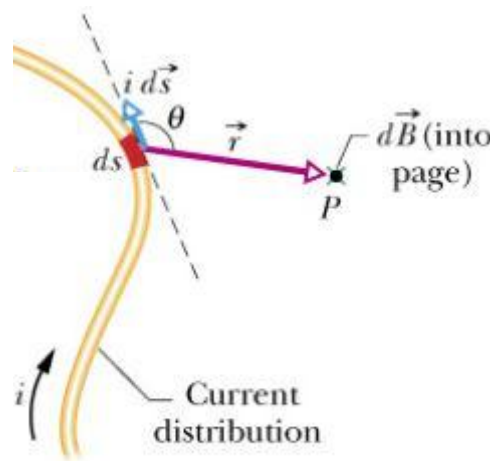
A wire bent into a semicircle of radius  $R$  forms a closed circuit and carries a current  $I$ . The wire lies in the  $xy$  plane, and a uniform magnetic field is directed along the positive  $y$  axis as in Figure below. Find the magnitude and direction of the magnetic force acting on the straight portion of the wire and on the curved portion.



## 2.2 SOURCES OF THE MAGNETIC FIELDS

### 2.2.1 The Biot-Savart Law

Let a certain conductor be carrying a current ' $I$ ' in a direction shown on the figure. Let ' $P$ ' be the point where the magnetic field due to the wire is to be studied. Let a small portion be considered which is of length ' $ds$ '. Let the line joining ' $ds$ ' and point ' $P$ ' from an angle  $\theta$  with the tangent to ' $ds$ '. Since the portion considered is very small, the magnetic field given by it at point  $P$  will also be small.



# The Biot-Savart Law

By experimental observations and empirically also,  $d\vec{B}$  is found to depend on several factors.

- The vector  $d\vec{B}$  is perpendicular both to  $d\vec{s}$  (which points in the direction of the current) and to the unit vector  $\hat{r}$  directed from  $d\vec{s}$  to  $P$ .
- The magnitude of  $d\vec{B}$  is inversely proportional to  $r^2$ , where  $r$  is the distance from  $d\vec{s}$  to  $P$ .
- The magnitude of  $d\vec{B}$  is proportional to the current and the magnitude  $ds$  of the length element  $d\vec{s}$ .
- The magnitude of  $d\vec{B}$  is proportional to  $\sin \theta$ , where  $\theta$  is the angle between the vectors  $d\vec{s}$  and  $\hat{r}$ .

# The Biot-Savart Law

These observations are summarized in the mathematical formula known today as the Biot-Savart law:

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{Id\vec{s} \times \hat{r}}{r^2}$$

where  $\mu_0$  is a constant called *permeability of free space*:

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

# The Biot-Savart Law

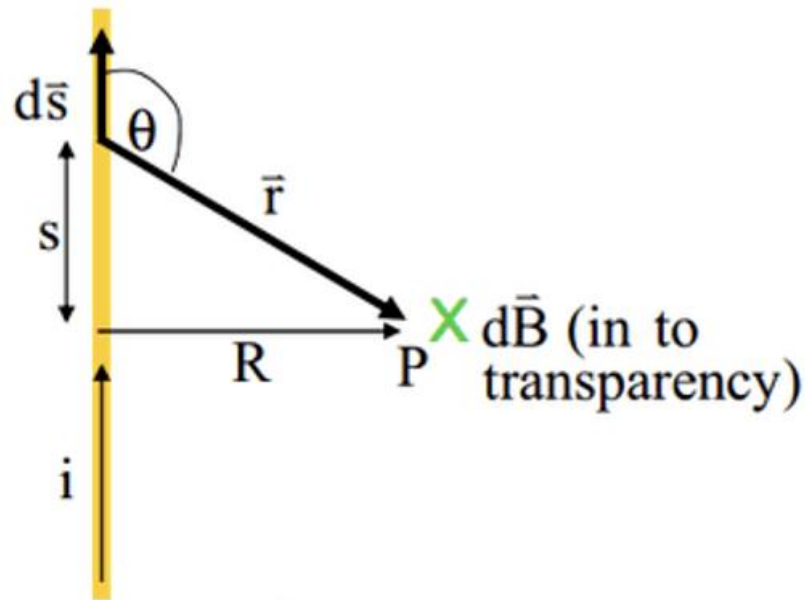
The overall magnetic field will be equal to the sum of all these small magnetic fields.

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{s} \times \hat{r}}{r^2}$$

With the integral taken over the entire current distribution.

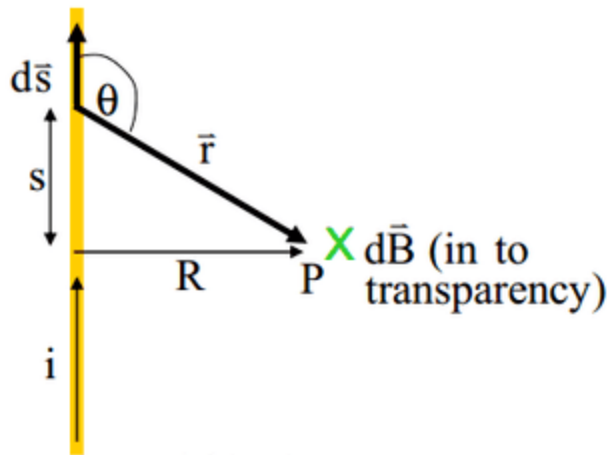
## 2.2.2. Magnetic Field Surrounding a Thin, Straight Conductor

- Consider long straight wire:



## 2.2.2. Magnetic Field Surrounding a Thin, Straight Conductor

- Consider long straight wire:



- $d\vec{B} = \frac{\mu_0}{4\pi} \frac{i d\vec{s} \times \vec{r}}{r^3}$

- $dB = \frac{\mu_0}{4\pi} \frac{i ds \sin \theta}{r^2}$

- Integrate:  $B = \int_{-\infty}^{\infty} dB = 2 \int_0^{\infty} dB = \frac{\mu_0 i}{2\pi} \int_0^{\infty} \frac{\sin \theta}{r^2} ds$

- Using the relationships  $r^2 = s^2 + R^2$  and

$$\sin \theta = \sin(\pi - \theta) = \frac{R}{\sqrt{s^2 + R^2}}$$

- We get:  $B = \frac{\mu_0 i}{2\pi} \int_0^{\infty} \frac{R}{(s^2 + R^2)^{\frac{3}{2}}} ds$

- Using the result:

$$\int \frac{dx}{(x^2 + a^2)^{\frac{3}{2}}} = \frac{x}{a^2 (x^2 + a^2)^{\frac{1}{2}}}$$

- The integral becomes:

$$B = \frac{\mu_0 i}{2\pi R} \frac{s}{\sqrt{s^2 + R^2}} \Bigg|_0^{\infty}$$

$$B = \frac{\mu_0 i}{2\pi R}$$

## 2.2.3 The Magnetic Force Between Two Parallel Conductors

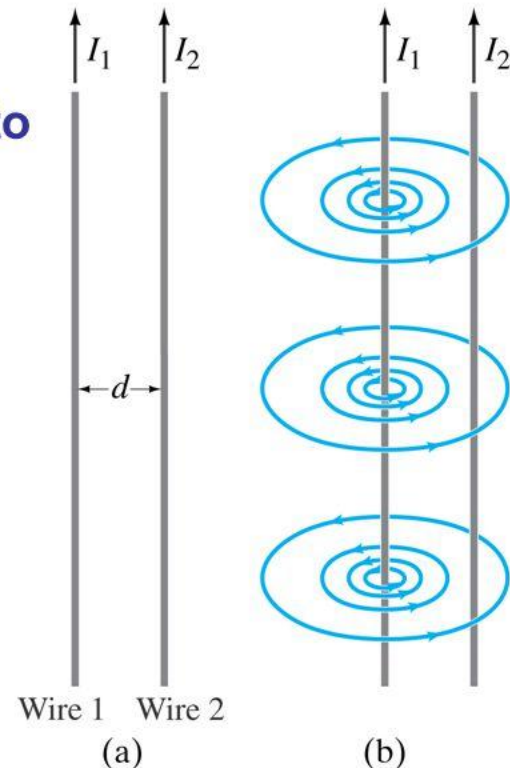
### Force between Two Parallel Wires

The magnetic field produced at the position of wire 2 due to the current in wire 1 is:

$$B_1 = \frac{\mu_0}{2\pi} \frac{I_1}{d}$$

The force this field exerts on a length  $l_2$  of wire 2 is:

$$F_2 = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d} l_2 \quad (20-7)$$

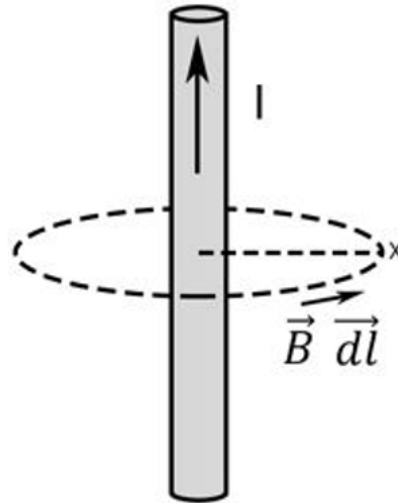


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# Ampere's Law

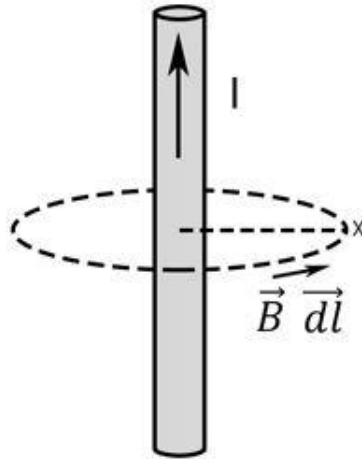
## Ampere's Law

states that for any closed loop path, the sum of the length elements times the magnetic field in the direction of the length element is equal to the permeability times the electric current enclosed in the loop.



# Ampere's Law

**Ampere's Law** states that for any closed loop path, the sum of the length elements times the magnetic field in the direction of the length element is equal to the permeability times the electric current enclosed in the loop.



$I$  : Electric current

$B$  : Magnetic field

$\mu_o$  : Permeability of free space

From Biot-Savart's Law, the magnetic field due to a long straight wire is,

$$B = \frac{\mu_o I}{2\pi r}$$

Since,  $\vec{B}$  and  $\vec{dl}$  are in the same direction,

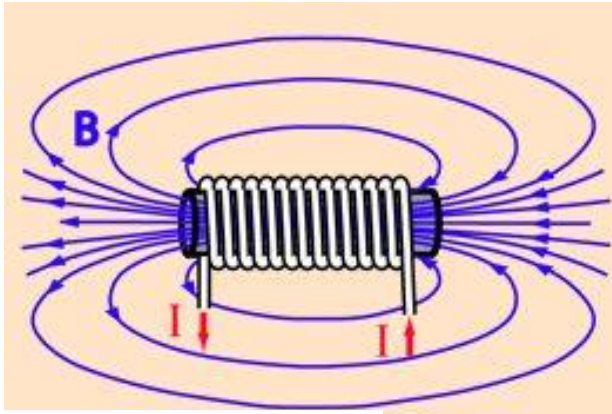
$$\vec{B} \cdot \vec{dl} = B \, dl \cos 0 = B \, dl$$

Therefore, 
$$\oint \vec{B} \cdot \vec{dl} = \frac{\mu_o I}{2\pi r} \oint dl$$

$$\Rightarrow \oint \vec{B} \cdot \vec{dl} = \frac{\mu_o I}{2\pi r} (2\pi r)$$

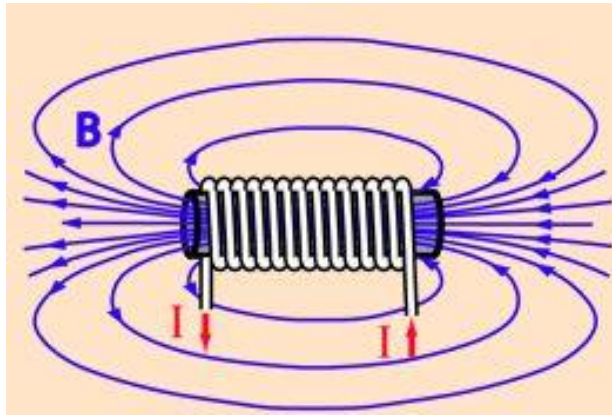
$$\boxed{\Rightarrow \oint \vec{B} \cdot \vec{dl} = \mu_o I}$$

## 2.2.6. Ampere's Law applications (**The magnetic field inside the solenoid** )



A solenoid is a long wire wound into a succession of closely spaced loops

## 2.2.6. Ampere's Law applications (**The magnetic field inside the solenoid** )



A solenoid is a long wire wound into a succession of closely spaced loops

**The magnetic field inside the solenoid is**

$$B = \frac{\mu_0 NI}{l}$$

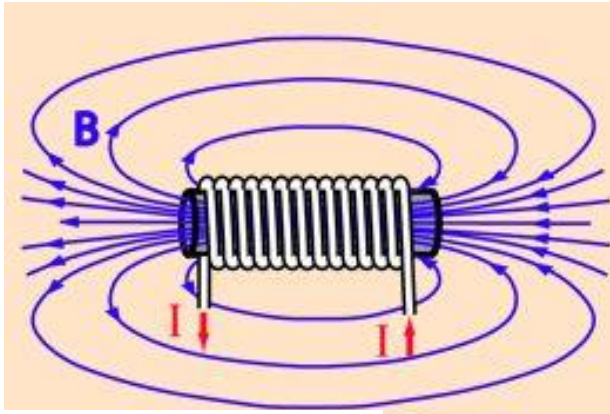
**Where**

**L-** The length of the coil

**I-** Current in the coil

**N-** Number of turns (loops) in solenoid

## 2.2.6. Ampere's Law applications (**The magnetic field inside the solenoid**)



A solenoid is a long wire wound into a succession of closely spaced loops

The magnetic field inside the solenoid is

$$B = \frac{\mu_0 N I}{l}$$

Where

**L**- The length of the coil

**I**- Current in the coil

**N**- Number of turns (loops) in solenoid

### Example

A solenoid has 15.0 turns per centimeter. What current will produce a magnetic field of  $3.00 \times 10^{-2}$  T within the solenoid?  $I = ?$

Solenoid B. Field

$$B = \mu_0 n I$$

$$B = \mu_0 n I$$

$$\Rightarrow I = \frac{B}{\mu_0 n}$$

$$= \frac{(3.00 \times 10^{-2} \text{ T})}{(1.26 \times 10^{-6} \text{ Tm/A})(\underline{15.0 \times 100})}$$

$$n = 15 \text{ turns/cm}$$

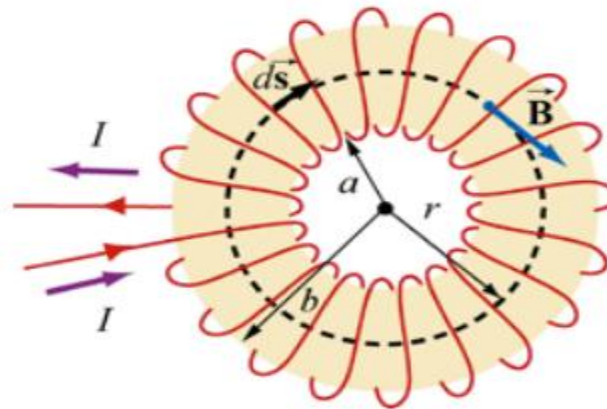
$$B = 3.00 \times 10^{-2} \text{ T}$$

$$\mu_0 = 1.26 \times 10^{-6} \text{ Tm/A}$$

$$= \boxed{15.9 \text{ A}}$$

### 2.2.6. Ampere's Law applications (The Magnetic Field of a Toroid (a circular solenoid))

Consider a toroid that consists of  $N$  turns, with inner radius  $a$  and outer radius  $b$ , as shown in Figure below. Find the magnetic field everywhere.



Applying Ampere's law using a circular loop of radius  $r$ , for the region  $a < r < b$ , we obtain

$$\oint \vec{B} \cdot d\vec{s} = \oint B ds = B \oint ds = B 2\pi r = \mu_0 N I$$

The magnitude of the magnetic field is

$$B = \frac{\mu_0 N I}{2\pi r} ; a < r < b$$

where  $r$  is the distance measured from the center of the toroid.

## 2.2.8 Magnetic Flux

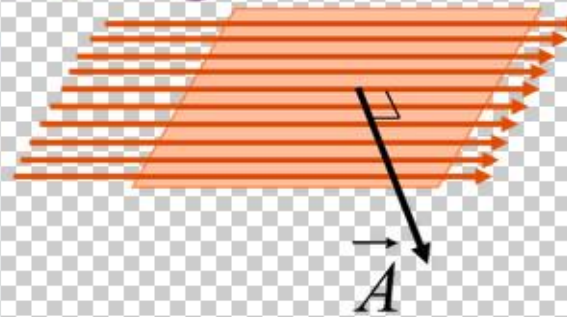
**Magnetic Flux,  $\Phi$ :** a measure of the number of magnetic field lines that cross a given area

- is defined as **the scalar product between the magnetic flux density,  $B$  and the vector of the surface area,  $A$**

$$\Phi = \vec{B} \bullet \vec{A} = BA \cos \theta$$

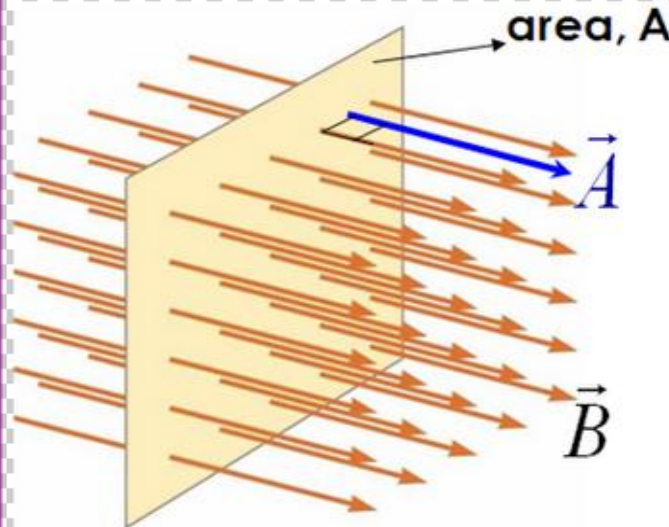
where,  
 $\Phi$ : magnetic flux  
 $\phi$ : angle between  $\vec{B}$  and  $\vec{A}$   
 $B$ : magnetic flux density  
 $A$ : area of the coil

- Scalar quantity**
- Unit:  **$T m^2$  or  $Wb$**



Direction of vector  $A$  always **perpendicular** to the surface area,  $A$ .

$$\theta = 90^\circ$$



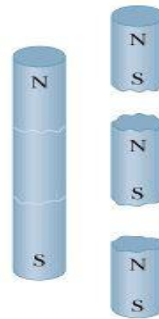
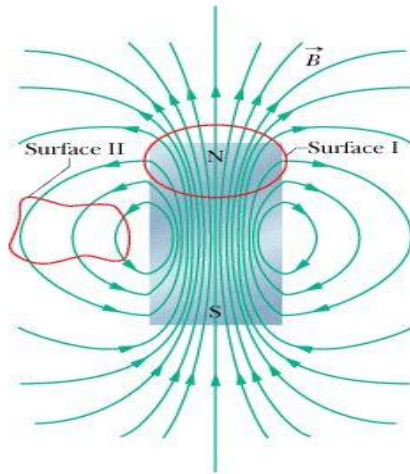
$$\theta = 0^\circ$$

$$\Phi = BA$$

# Gauss' Law for Electric and Magnetic Fields

We studied Gauss' law for electric fields:

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$



The simplest magnetic structure that can exist is a magnetic dipole. Magnetic monopoles do not exist (as far as we know).

Gauss' law for magnetic fields is a formal way of saying that magnetic monopoles do not exist. The law asserts that the net magnetic flux  $\Phi_B$  through any closed Gaussian surface is zero:

$$\Phi_B = \oint \vec{B} \cdot d\vec{A} = 0$$

- Magnetic fields do not begin or end at any point
  - The number of lines entering a surface equals the number of lines leaving the surface

Because magnetic field lines are continuous loops, all closed surfaces have as many magnetic field lines going in as coming out. Hence, the net magnetic flux through a closed surface is zero.

## 2.3. Faraday's Law of induction

- **Faraday's law of induction:** the instantaneous emf induced in a circuit is directly proportional to the time **rate** of change of the magnetic flux through the circuit
- If the circuit consists of  $N$  loops, all of the same area, and if  $\Phi_B$  is the flux through one loop, an emf is induced in every loop and Faraday's law becomes

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

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The minus sign denotes Lenz's Law. Emf is the term for generated or induced voltage.

$$\mathcal{E} = - \frac{d\Phi_B}{dt}$$

### *Definition:*

*Induced EMF, also known as electromagnetic induction or EMF Induction is the production of voltage in a coil because of the change in a magnetic flux through a coil.*

# LENZ LAW

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*“An induced current always flows in a direction such that it opposes the change which produced it”. See the figures below*

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This just says that **‘the direction of the Induced Current is such that it opposes the change that causes it’.**

$$V = - \frac{dN\Phi}{dt}$$

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When the "N" Pole of the magnet is moved towards the coil, end of the coil becomes "N" Pole.

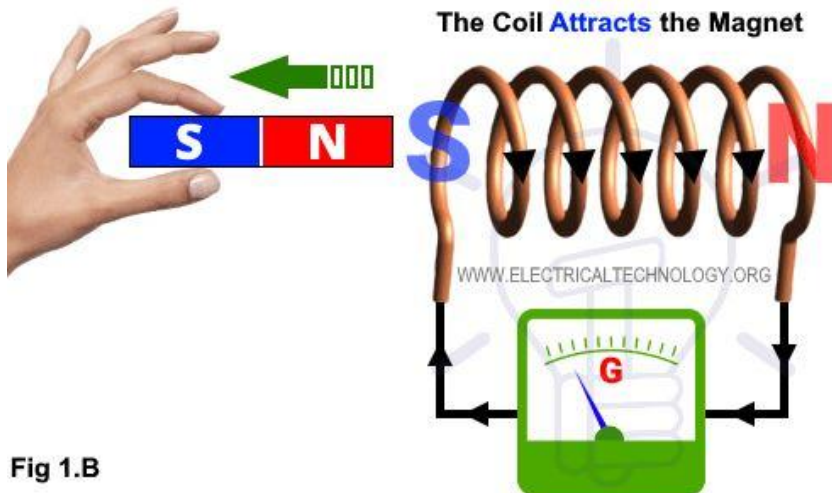


Fig 1.B

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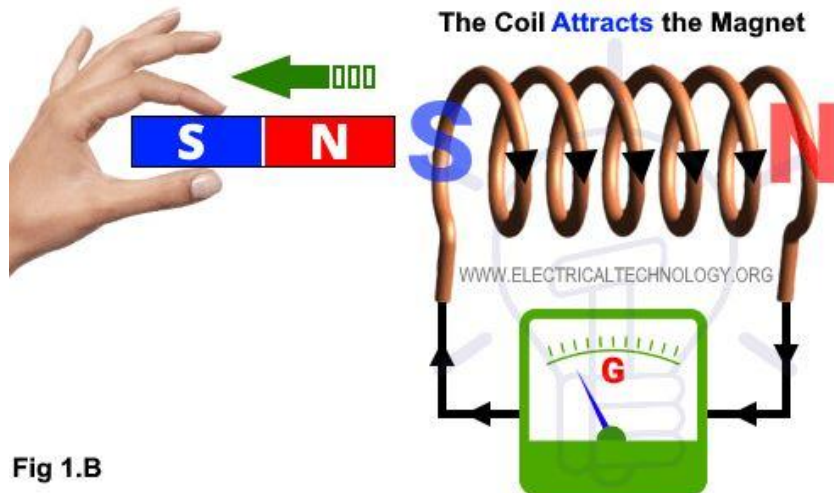


Fig 1.B

When the "N" Poles of the magnet is moved away from the coil, end of the coil becomes "S" Pole.

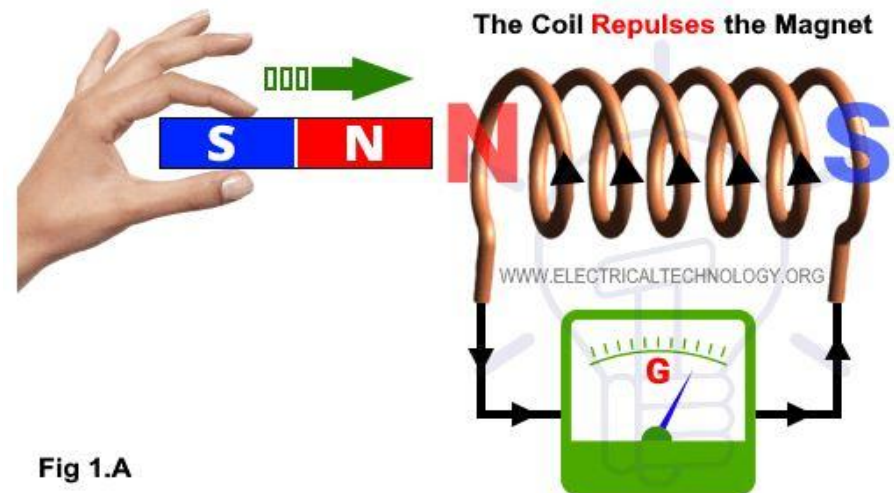


Fig 1.A

# Self inductance

**Self Inductance, L:** the ratio of the self induced e.m.f. to the rate of change of current in the coil

$$L = -\frac{\varepsilon}{dI/dt} \Rightarrow \varepsilon = -L \frac{dI}{dt}$$

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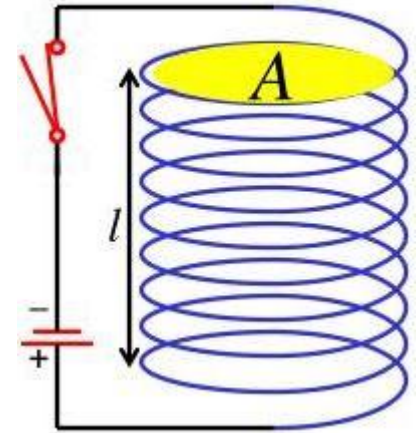
$$L = -\frac{\varepsilon}{dI/dt} \quad \Rightarrow \quad \varepsilon = -L \frac{dI}{dt}$$

The inductance is measured in Henrys

Consider a solenoid of length  $l$ , area  $A$  and  $N$  turns, connected to a battery of emf  $\mathcal{E}$  (or  $E$ ).

What happens as you close the switch?

- Lenz Law– loops resists change in magnetic field
- Magnetic field caused by the current
- “Inductor” resists change in current



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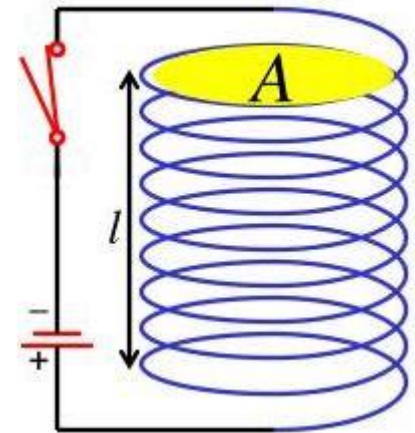
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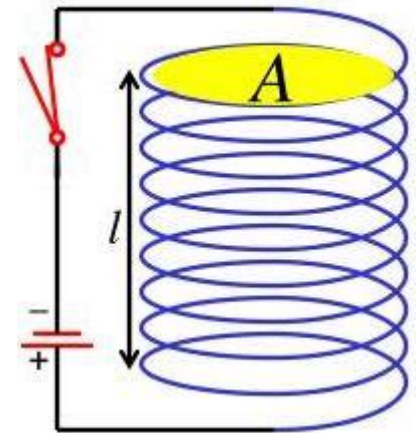
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$$E = -L \frac{dI}{dt} \quad L = \mu_0 N^2 A / \ell$$

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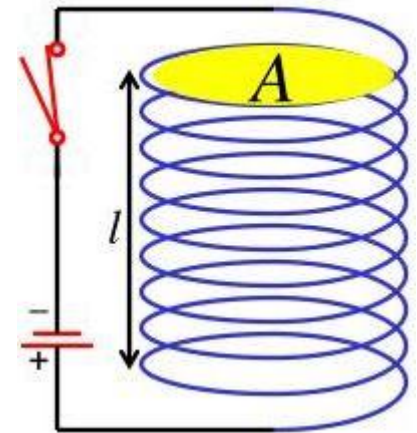
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Self Inductance of a solenoid

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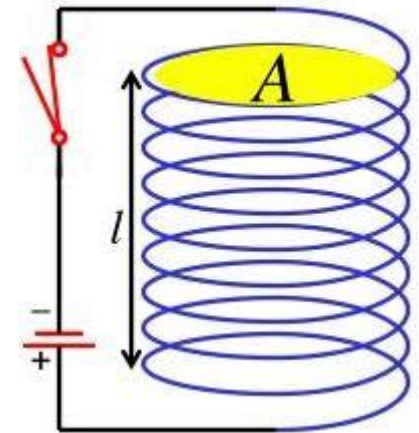
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**NB:** The self-inductance of a toroid in the limit of  $l=2\pi a$  has the same form as that of a solenoid

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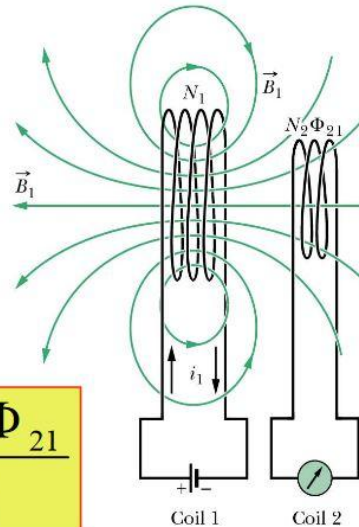
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# Mutual inductance

- **Mutual induction** – current in one coil induces emf in other coil
- Distinguish from **self-induction**
- Mutual inductance,  $M_{21}$  of coil 2 with respect to coil 1 is

$$L = \frac{N\Phi_B}{i}$$

$$M_{21} = \frac{N_2 \Phi_{21}}{i_1}$$

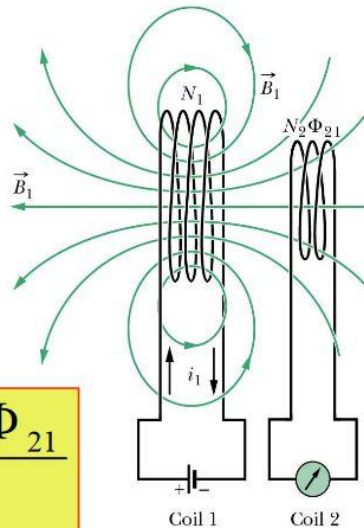


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- Rearrange equation

$$M_{21} i_1 = N_2 \Phi_{21}$$

- Vary  $i_1$  with time

$$M_{21} \frac{di_1}{dt} = N_2 \frac{d\Phi_{21}}{dt}$$

- Faraday's law

$$\mathcal{E}_2 = -N_2 \frac{d\Phi_{21}}{dt}$$

- Induced emf in coil 2 due to  $i$  in coil 1 is

$$\mathcal{E}_2 = -M_{21} \frac{di_1}{dt}$$

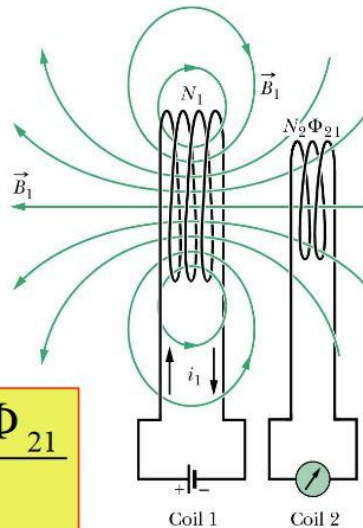
- Obeys Lenz's law (minus sign)

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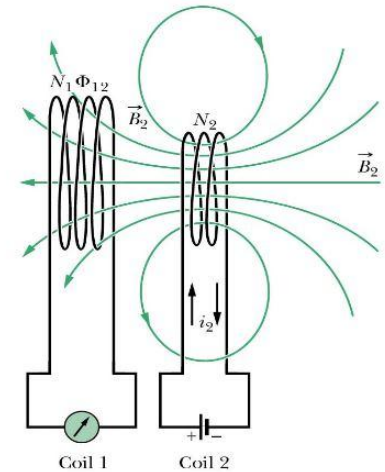
$$M_{21} = \frac{N_2 \Phi_{21}}{i_1}$$



- Reverse roles of coils
- What is induced emf in coil 1 from a changing current in coil 2?
- Same game as before

$$M_{12} = \frac{N_1 \Phi_{12}}{i_2}$$

$$\mathcal{E}_1 = -M_{12} \frac{di_2}{dt}$$



$$M_{21} = \frac{N_2 \Phi_{21}}{i_1}$$

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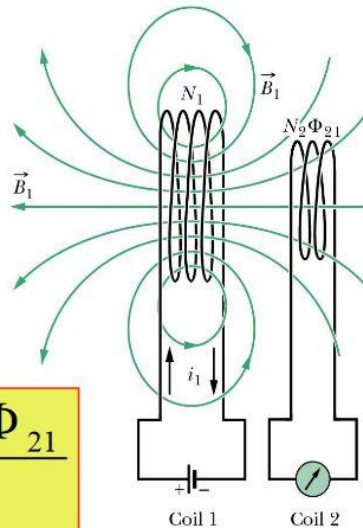
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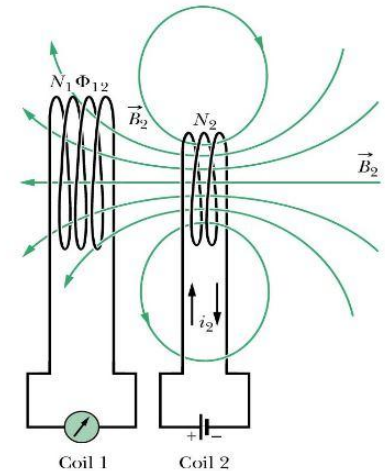
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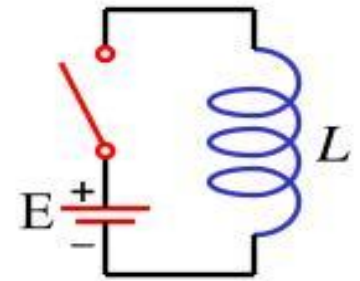
- The mutual inductance terms are equal

$$M_{12} = \frac{N_1 \Phi_{12}}{i_2}$$

$$M_{21} = \frac{N_2 \Phi_{21}}{i_1}$$

$$M_{21} = M_{12} = M$$

# Energy in Inductors



- Is the battery doing work on the inductor?

$$P = I(\Delta V) = IL \frac{dI}{dt}$$

- Integral of power is work done on the inductor

$$U = \int P dt = \int IL \frac{dI}{dt} dt = L \int I dI = \frac{1}{2} LI^2 + k$$

- It makes sense to say there is no energy in inductor with no current

$$U = \frac{1}{2} LI^2$$

$$L = \mu_0 N^2 A / \ell$$

$$B = \frac{\mu_0 NI}{\ell}$$

- Energy density inside a solenoid?

$$U = \frac{\mu_0 N^2 A I^2}{2\ell}$$

$$u = \frac{U}{A\ell} = \frac{\mu_0 N^2 I^2}{2\ell^2} = \frac{B^2}{2\mu_0}$$

- Just like with electric fields, we can associate the energy with the magnetic fields, not the current carrying wires

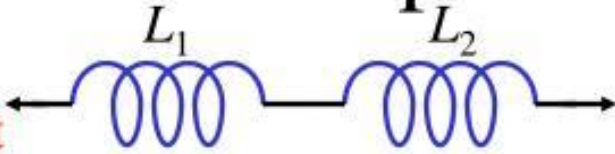
$$u = \frac{B^2}{2\mu_0}$$

# Inductors in series and parallel

•For inductors in series, the inductors have the same current

•Their EMF's add

$$L = L_1 + L_2$$

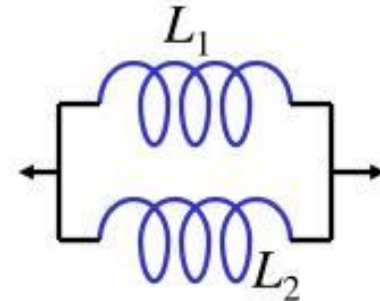


$$E = -L_1 \frac{dI}{dt} - L_2 \frac{dI}{dt} = -(L_1 + L_2) \frac{dI}{dt}$$

•For inductors in parallel, the inductors have the same EMF but different currents

$$E = -L_1 \frac{dI_1}{dt}$$

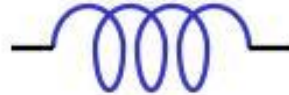
$$E = -L_2 \frac{dI_2}{dt}$$



$$\frac{dI}{dt} = \frac{dI_1}{dt} + \frac{dI_2}{dt} = -\frac{E}{L_1} - \frac{E}{L_2} = -\frac{E}{L}$$

$$\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2}$$

# Parallel and Series - Formulas

|                     | Capacitor<br> | Resistor<br> | Inductor<br> |
|---------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Series              | $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$                                                  | $R = R_1 + R_2$                                                                                 | $L = L_1 + L_2$                                                                                 |
| Parallel            | $C = C_1 + C_2$                                                                                | $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$                                                   | $\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2}$                                                   |
| Fundamental Formula | $\Delta V = \frac{Q}{C}$                                                                       | $\Delta V = IR$                                                                                 | $E_L = -L \frac{dI}{dt}$                                                                        |

# MAXWELL'S EQUATIONS

## Maxwell's laws

- Basis of all electrical and magnetic phenomena can be described by 4 equations called **Maxwell's equations**
- As fundamental to electromagnetism as Newton's law are to mechanics
- Einstein showed that Maxwell's equations work with special relativity

## Maxwell's 4 equations are

- Gauss' Law  $\oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$
- Gauss' Law for magnetism  $\oint \vec{B} \cdot d\vec{A} = 0$
- Faraday's Law  $\oint \vec{E} \cdot d\vec{s} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A}$
- Ampere-Maxwell Law  $\oint \vec{B} \cdot d\vec{s} = \epsilon_0 \mu_0 \frac{d}{dt} \int \vec{E} \cdot d\vec{A} + \mu_0 i$

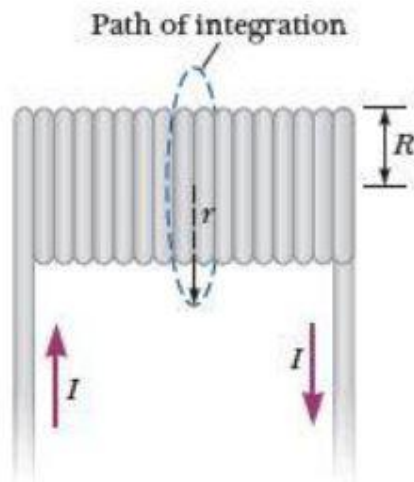
## Maxwell's laws in differential form

- Gauss' Law  $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}; \quad \rho = \text{charge density}$
- Gauss' Law for magnetism  $\vec{\nabla} \cdot \vec{B} = 0$
- Faraday's Law  $\vec{\nabla} \times \vec{E} = -\frac{d\vec{B}}{dt}$
- Ampere-Maxwell Law  $\vec{\nabla} \times \vec{B} = \frac{1}{c^2} \frac{d\vec{E}}{dt} + \mu_0 \vec{J}$   
 $c^2 = \frac{1}{\mu_0 \epsilon_0}; \quad \vec{J} = \text{current density}$

# Example 4

A long solenoid of radius  $R$  has  $n$  turns of wire per unit length and carries a time varying current that varies sinusoidally as  $I_{\max} \cos \omega t$ , where  $I_{\max}$  is the maximum current and  $\omega$  is the angular frequency of the alternating current source (Figure below).

- a) Determine the magnitude of the induced electric field outside the solenoid at a distance  $r > R$  from its long central axis.
- b) What is the magnitude of the induced electric field inside the solenoid, a distance  $r$  from its axis?

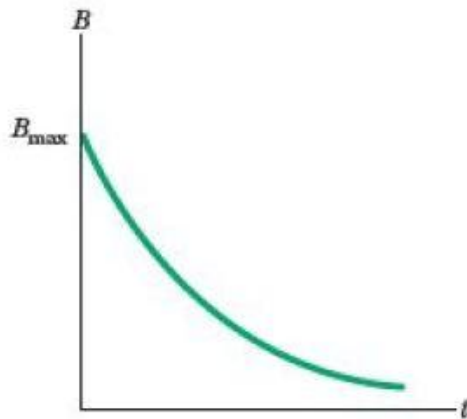


## Example 5

A coil consists of **200 turns** of wire. Each turn is a square of side  **$d = 18 \text{ cm}$** , and a uniform magnetic field directed perpendicular to the plane of the coil is turned on. If the field changes linearly from **0** to  **$0.50 \text{ T}$**  in  **$0.80 \text{ s}$** , what is the magnitude of the induced **emf** in the coil while the field is changing?

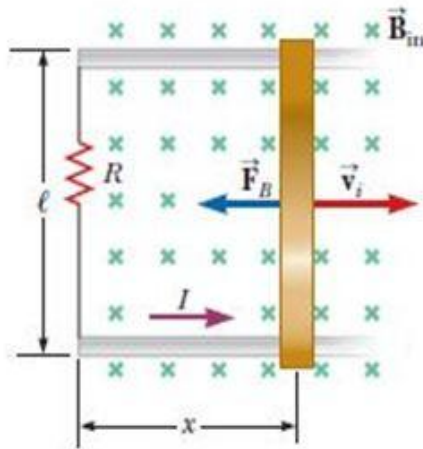
# QUIZ 2

A loop of wire enclosing an area  $A$  is placed in a region where the magnetic field is perpendicular to the plane of the loop. The magnitude of  $\vec{B}$  varies in time according to the expression  $B = B_{\max}e^{-at}$ , where  $a$  is some constant. That is, at  $t = 0$ , the field is  $B_{\max}$ , and for  $t > 0$ , the field decreases exponentially (Figure below). Find the induced **emf** in the loop as a function of time. Evaluate the maximum **emf**



# Example 6

The conducting bar illustrated in the Figure below, moves on two frictionless, parallel rails in the presence of a uniform magnetic field directed into the page. The bar has mass  $m$ , and its length is  $\ell$ . The bar is given an initial velocity  $\mathbf{v}_i$  to the right and is released at  $t = 0$ . Using Newton's laws, find the velocity of the bar as a function of time.



# Exercise 1

A toroidal solenoid with mean radius  $r$  and cross-sectional area  $A$  is wound uniformly with  $N_1$  turns. A second toroidal solenoid with  $N_2$  turns is wound uniformly on the top of the first, so that the two solenoids have the same cross-sectional area and mean radius. (a) What is the mutual inductance of the two solenoids? Assume that the magnetic field of the first solenoid is uniform across the cross section of the two solenoids. (b) If  $N_1=500$  turns,  $N_2=300$  turns,  $r=10.00$  cm, and  $A=0.800$  cm<sup>2</sup>, what is the value of the mutual inductance?

# Exercise2

An inductor with an inductance  $2.50 \text{ H}$  and a resistance of  $8.00 \Omega$  is connected to the terminals of a battery with an emf of  $6.00 \text{ V}$  and negligible internal resistance. Find (a) the initial rate of increase of current in the circuit; (b) the rate of increase of current at the instant when the current is  $0.500 \text{ A}$ ; (c) the current  $0.250 \text{ s}$  after the circuit is closed; (d) the final steady-state current.

# Exercise3

A  $20.0\ \mu\text{F}$  capacitor is charged by a  $150.0\text{-V}$  power supply, then disconnected from the power and connected in series with a  $0.280\text{-mH}$  inductor. Calculate: (a) the oscillation frequency of the circuit; (b) the energy stored in the capacitor at time  $t=0\text{ ms}$  (the moment of connection with the inductor); (c) the energy stored in the inductor at  $t=1.30\text{ ms}$ .

# Exercise4

A capacitor with capacitance  $6.00 \times 10^{-5} \text{ F}$  is charged by connecting it to a 12.0-V battery. The capacitor is disconnected from the battery and connected across an inductor  $L=1.50 \text{ H}$ . (a) What are the angular frequency of the electrical oscillations and the period of these oscillations (the time for one oscillation)? (b) What is the initial charge on the capacitor? (c) How much energy is initially stored in the capacitor? (d) What is the charge on the capacitor 0.0230 s after the connection to the inductor is made? Interpret the sign of your answer. (e) At the time given in part (d), what is the current in the inductor? Interpret the sign of your answer. (f) At the time given in part (d), how much electrical energy is stored in the capacitor and how much is stored in the inductor?

# Exercise 5

An  $L$ - $R$ - $C$  series circuit has  $L = 0.450\text{ H}$ ,  $C = 2.50 \times 10^{-5}\text{ F}$ , and resistance  $R$ , (a) What is the angular frequency of the circuit when  $R=0$ ? (b) What value must  $R$  have to give a 5.0% decrease in angular frequency compared to the value calculated in part (a)?

***END OF THE MODULE,  
Good Luck In Your  
Exams***